

Chapter-03

1. Basic Terminology

Before learning encryption, we need to understand some important terms.

Plaintext

- The **original readable message** before encryption.
- It is the information that the sender wants to protect.

Example: HELLO WORLD

This is plaintext because anyone can read it.

Ciphertext

- The encrypted or coded message.
- It cannot be understood without the correct key.

Example

KHOOR ZRUOG

This is ciphertext.

Cipher

A cipher is an algorithm or mathematical procedure that converts plaintext into ciphertext and vice versa.

Example:

Caesar Cipher shifts every letter by 3 positions.

Key

A key is a secret value used by the encryption algorithm. Only the sender and receiver should know the key.

Example

If the Caesar Cipher uses

Key = 3

every letter is shifted by 3 positions.

Encryption (Encipher)

Encryption means converting plaintext into ciphertext.

Formula

Plaintext → Ciphertext

Example

HELLO

↓

KHOOR

Decryption (Decipher)

Decryption means converting ciphertext back into plaintext.

Formula

Ciphertext → Plaintext

Example

KHOOR

↓

HELLO

Cryptography

Cryptography is the science of protecting information by using encryption techniques. Its main goal is to provide secure communication.

Cryptanalysis

Cryptanalysis is the study of breaking encryption without knowing the secret key. It is also called code breaking. The goal is to recover the plaintext or the secret key.

Cryptology

Cryptology is the science of protecting and studying secret information. It involves techniques to keep data safe from unauthorized people and methods to test whether those protections are secure.

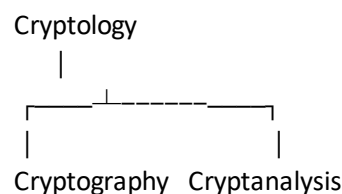
Cryptology has two main parts:

- **Cryptography** – The process of protecting information by converting readable data (plaintext) into an unreadable form (ciphertext) using encryption, so that only authorized people can read it.
- **Cryptanalysis** – The process of analyzing and breaking encrypted information to find the original message or discover weaknesses in the encryption system without knowing the secret key.

Simple Formula

Cryptology = Cryptography + Cryptanalysis

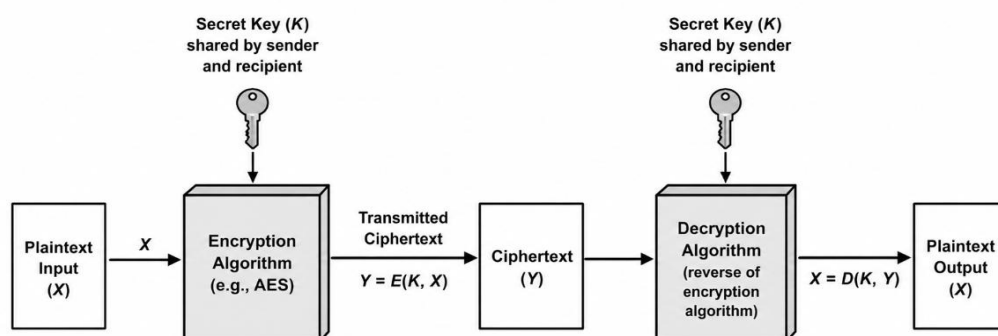
- **Cryptography** → Makes information secure.
- **Cryptanalysis** → Tests or tries to break that security.



2. Symmetric Cipher Model

A Symmetric Cipher uses one secret key for both encryption and decryption. The sender and receiver must both know the same key.

Working Process



Mathematical Representation

Encryption

$$Y = E(K, X)$$

where

- X = Plaintext
- K = Secret Key
- E = Encryption Algorithm
- Y = Ciphertext

Decryption

$$X = D(K, Y)$$

where

- D = Decryption Algorithm

Requirements for Secure Symmetric Encryption

A secure symmetric cipher needs two requirements.

1. Strong Encryption Algorithm

The algorithm should be difficult to break even if everyone knows how it works. The security should depend on the **key**, not on hiding the algorithm.

2. Secret Key

The sender and receiver must share a secret key. If an attacker gets the key, the encrypted message becomes readable.

Key Distribution Problem

The sender and receiver need a secure communication channel to exchange the secret key. This is one of the biggest challenges in symmetric encryption.

3. Cryptography

A cryptographic system can be classified in three ways.

A. Based on Encryption Operation

1. Substitution

Replace one letter (or bit) with another.

Example

A → D

B → E

C → F

Example cipher

- Caesar Cipher
- Monoalphabetic Cipher

2. Transposition

The letters remain the same, but their positions are rearranged.

Example

HELLO

↓

LHOEL

3. Product Cipher

A combination of

- Substitution
- Transposition

Modern encryption algorithms (such as AES) use product ciphers.

B. Based on Number of Keys

Single-Key (Private-Key)

Uses **one secret key**.

Example: AES, DES

Two-Key (Public-Key)

Uses

- Public Key
- Private Key

Example: RSA

C. Based on Plaintext Processing

Block Cipher

Encrypts fixed-size blocks.

Example

128 bits

↓

Encrypted together

Example: AES

Stream Cipher

Encrypts one bit or one character at a time.

Example

HELLO

↓

H

↓

E

↓

L

↓

L

↓

O

4. Cryptanalysis

Cryptanalysis is the process of breaking an encryption system. The main goal is not only to read the message but also to discover the secret key. Once the key is found, every message encrypted with that key becomes readable.

Two General Approaches

1. Cryptanalytic Attack

Uses

- mathematics
- statistics
- language properties to recover the key.

2. Brute Force Attack

Simply tries every possible key until the correct one is found.

5. Types of Cryptanalytic Attacks

1. Ciphertext-Only Attack

The attacker knows

- encryption algorithm
- ciphertext

The attacker does not know the plaintext. They try to guess the message using statistical analysis.

2. Known-Plaintext Attack

The attacker knows

- some plaintext
- corresponding ciphertext

Using this information, they try to discover the secret key.

3. Chosen-Plaintext Attack

The attacker chooses any plaintext, gets it encrypted, then studies the resulting ciphertext.

4. Chosen-Ciphertext Attack

The attacker chooses ciphertext, gets it decrypted, and studies the resulting plaintext.

5. Chosen-Text Attack

The attacker can choose

- plaintext
- ciphertext

for encryption or decryption. This is the most powerful attack model.

6. Security Definitions

Unconditional Security (Perfect Security)

A cipher is unconditionally secure (also called information-theoretically secure) if it cannot be broken, even if an attacker has unlimited computing power, unlimited time, and unlimited resources. The encrypted message does not provide enough information to determine the original plaintext without the secret key.

Key Points:

- Security does not depend on the attacker's computing power.
- Even trying every possible key will not reveal the correct plaintext unless the secret key is known.
- Provides perfect secrecy.

Example:

- One-Time Pad (OTP) – When the key is truly random, as long as the message, used only once, and kept secret, it is mathematically impossible to break.

Computational Security (Practical Security)

A cipher is computationally secure if it can theoretically be broken, but doing so would require an impractical amount of time, computing power, or resources. In practice, the attack is so difficult that it is considered impossible.

Key Points:

- Security depends on the limitations of current and future computers.
- The cipher is not mathematically unbreakable, but breaking it is computationally infeasible.
- Most modern encryption algorithms rely on computational security.

Example:

- AES (Advanced Encryption Standard) with a 256-bit key.
- RSA with sufficiently large key sizes (e.g., 2048 bits or higher).
- If breaking a key would take 10^{20} years using the fastest available computers, the cipher is considered computationally secure.

7. Brute Force Search

A brute-force attack tries every possible key until the correct one is found. It is the simplest attack.

Characteristics

- Guaranteed to work eventually.
- No knowledge of encryption is required.
- Larger key sizes require much more time.

8. Classical Substitution Cipher

Definition

A substitution cipher replaces each plaintext letter with another letter, number, or symbol. Example

HELLO

↓

KHOOR

9. Monoalphabetic Cipher

A Monoalphabetic Cipher uses one substitution alphabet. Every plaintext letter always maps to the same ciphertext letter.

Example

Plain

A → D

B → K

C → V

D → Q

The mapping never changes.

Example

Plain Alphabet

ABCDEFGHIJKLMNOPQRSTUVWXYZ

Cipher Alphabet

DKVQFIBJWPESCXHTMYAUOLRGZN

Plaintext

IF WE WISH TO REPLACE LETTERS

Ciphertext

WIRFRWAJUHIFTSDVFSFUUFYA

Security

The number of possible keys is

$$26!$$

which is approximately

$$4 \times 10^{26}$$

This is an extremely large number.

However, Monoalphabetic Cipher is still insecure. Why? Because English language has predictable letter frequencies.

10. Caesar Cipher

The Caesar Cipher is a monoalphabetic substitution cipher in which each letter of the plaintext is replaced by another letter located a fixed number of positions forward or backward in the alphabet.

Example

Plaintext

MEET ME AFTER THE TOGA PARTY

Ciphertext

PHHW PH DIWHU WKH WRJD SDUWB

Alphabet Shift

Plain

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z

Cipher

D E F G H I J K L M N O P Q R S T U V W X Y Z A B C

Number Representation

A = 0

B = 1

...

Z = 25

Encryption Formula

$$C = (P + K) \bmod 26$$

where

- P = Plaintext
- K = Key
- C = Ciphertext

Decryption Formula

$$P = (C - K) \bmod 26$$

11. Cryptanalysis of Caesar Cipher

The Caesar cipher is one of the easiest encryption techniques to break because it has a very small key space.

How Caesar Cipher is Broken

- The Caesar cipher has only 26 possible keys (shifts).
- The letter A can be shifted to A, B, C, ..., Z, resulting in 26 possible ciphertexts.
- An attacker can simply try each possible shift one by one until the original message is found.
- This method is called a brute-force attack (exhaustive key search).

- The attacker must recognize which decrypted text is meaningful English (or the expected language).

12. Language Redundancy

Language redundancy means that human languages contain repeated patterns and predictable information, which helps in understanding messages even when some parts are missing.

Example:

"th lrd s m shphrd shll nt wnt" → "The Lord is my shepherd; I shall not want."

Letter Frequency Analysis

- Letters in a language do not appear equally.
- In English, the most common letters are:
E, T, R, N, I, O, A, S
- Rare letters include: **Z, J, K, Q, X**

Cryptanalysts use single, double, and triple letter frequency tables to identify patterns and break encryption systems.

13. English Letter Frequencies

Not all English letters occur equally often.

Approximate order (most frequent → least frequent):

Rank	Letters
High Frequency	E, T, A, O, I, N
Medium Frequency	S, H, R, D, L, C
Low Frequency	U, M, W, F, G, Y, P
Very Rare	B, V, K, J, X, Q, Z

Attackers use these frequency patterns to guess substitutions.

14. Using Letter Frequencies in Cryptanalysis

Monoalphabetic substitution does not change the frequency of letters.

Example: If **E** is the most common letter in plaintext, its encrypted version will also be the most common letter in ciphertext.

Therefore,

attackers:

1. Count letter frequencies.
2. Compare them with normal English frequencies.
3. Guess the substitutions.
4. Gradually recover the key.

This technique is called **frequency analysis**.

15. Example of Cryptanalysis

Suppose the ciphertext is:

UZQSOVUOHXMOPVGPOZPEVSGZWSZOPFPES...

Step 1

Count how many times each letter appears.

Suppose, P appears most often.

Step 2

Guess

P → E

because **E** is the most common English letter.

Step 3

Suppose, Z is the second most common.

Guess

$Z \rightarrow T$

Step 4

Suppose

ZW

appears frequently.

Guess

$ZW \rightarrow TH$

Then

$ZWP \rightarrow THE$

Step 5

Continue making educated guesses until the message becomes readable. Eventually the plaintext is recovered.